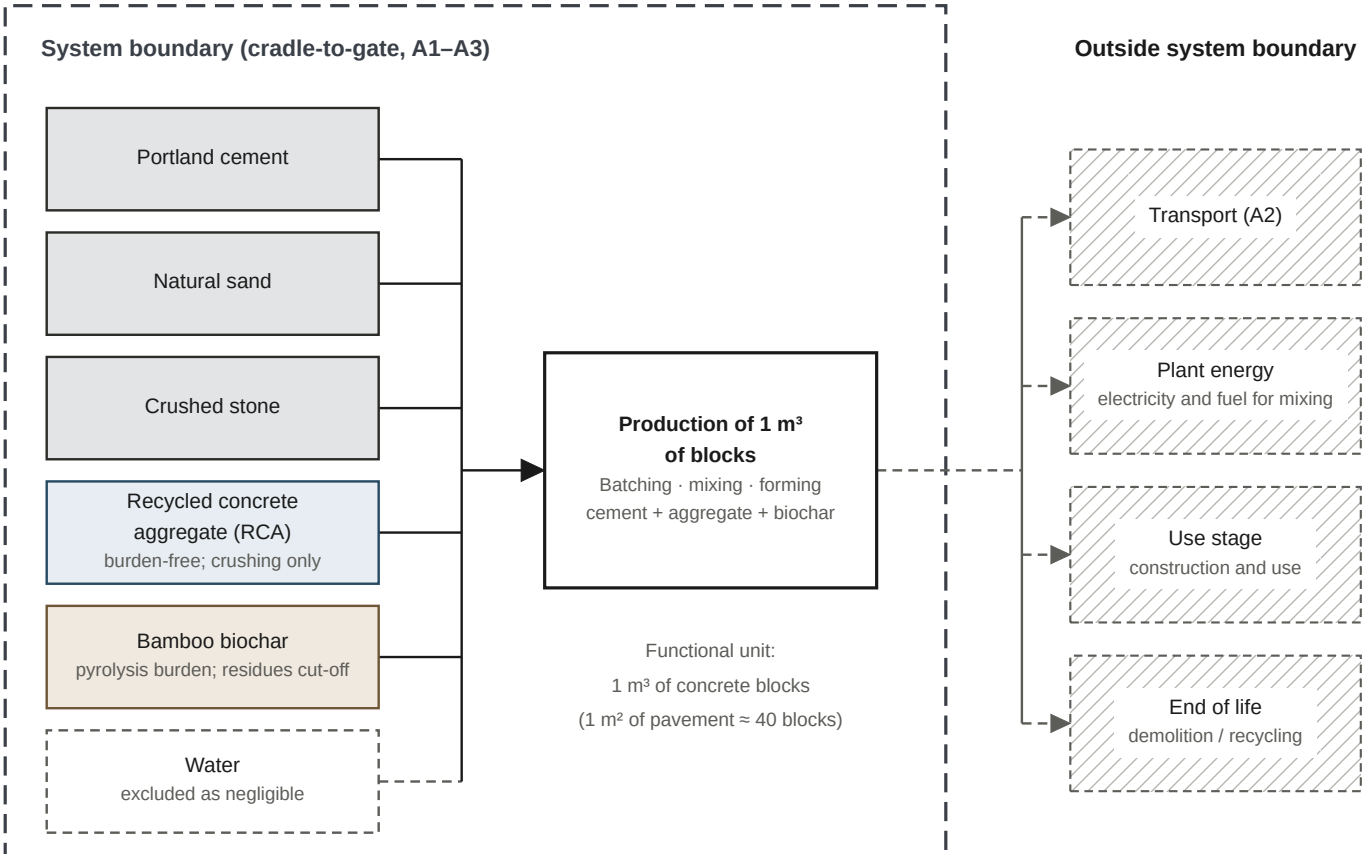
System boundary (cradle-to-gate, A1–A3)
Outside system boundary
Portland cement
Natural sand
Crushed stone
Recycled concrete aggregate (RCA)
burden-free; crushing only
Bamboo biochar
pyrolysis burden; residues cut-off
Water
excluded as negligible
Production of 1 m³ of blocks
Batching · mixing · forming
cement + aggregate + biochar
Functional unit:
1 m³ of concrete blocks
(1 m² of pavement ≈ 40 blocks)
Transport (A2)
Plant energy
electricity and fuel for mixing
Use stage
construction and use
End of life
demolition / recycling